Voyager 1 and 2
Atlas of Six Saturnian Satellites

NASA

APPENDIX B

The Saturnian System

The highly successful missions of the Pioneer and Voyager 1 and 2 (*Science,* 1980, 1981, 1982; Morrison, 1982) spacecraft have tremendously expanded our knowledge about the planet Saturn, its rings, and its satellites. The system has been found to be an extremely complex group of at least 17 satellites and thousands of rings orbiting a huge planet (120 000 km or 74 500 miles in diameter). The size of this system may be more fully appreciated if it is realized that the orbit of Earth's Moon would extend only to the outer limit of Saturn's ring system (Ring E) or that the largest moon, Titan, is almost half the diameter of Earth. Saturn itself is composed mostly of lighter gases such as hydrogen, deuterium, methane, ammonia, ethane, phosphene, and probably helium. Because of its rapid rotation (10 hr, 14 min), these gases are forced outward, resulting in an equatorial diameter 11 percent greater than the polar diameter. Because of its composition, the planet has a very low density; "average" Earth material weighs almost eight times as much as average Saturn material. Given a large enough body of water, we could float Saturn in it. Because of this low density and size, Saturn's gravity near the top of the atmosphere is only 0.9 times that of Earth; that is, anything that weighs 100 lb on Earth would weigh only 90 lb on Saturn. Temperatures in the outer atmosphere of Saturn range from −130° C (−265° F) to −180° C (−355° F). Ring particles and satellites appear to be frozen gases and water ice, and their densities indicate that the term "dirty snowball" is a reasonably accurate description for most of them. Ice at these low temperatures is as brittle as most rock, and eons of impacts have scarred the surfaces of the satellites with craters and massive fractures.

Saturn is the most distant planet known to the ancients, and its name is a Latin term for the Roman god of agriculture. From Earth, it is a bright object (magnitude −0.4 at greatest brilliancy) with a slightly orange color. More than 29 years (29.485 years to be exact) are required for this planet to complete one revolution about the Sun, and its path can be traced in our night sky as it slowly moves through the constellations of the ecliptic. In 1610, when Galileo first turned his crude telescope on Saturn, he was mystified by what he described as "companions" of Saturn. He drew Saturn as a planet with two handlelike protuberances. As telescopes were improved, the rings were resolved more clearly, and in 1655 Huygens finally perceived their geometry. In 1675, Cassini noted a "gap" in the rings, which today bears the name Cassini Division. Voyager found that this is not a gap at all, but a ring that appears from Earth to be darker than adjacent rings. The rings separated by this "division" were named Rings A and B by W. Struve. An additional division was found in Ring A by J. Encke in 1837 and is now called Encke's Division. A third ring inside Ring B was noted in 1838 by J. G. Galle and is known as the Crepe or Ring C. Other rings and ring features were suspected by observers, but were not resolved until spacecraft traversed the Saturnian system in 1980 and 1981.

The satellites of Saturn were discovered by painstaking observation over a period of 250 years. The moon Iapetus has been of great interest to astronomers because of its significant variations of brightness (magnitude variation of 1.92) at different positions in its orbit. In 1944, G. Kuiper noted the presence of a methane atmosphere on Titan, further expanding the mysteries of the Saturnian system. The satellites are all fairly substantial

Table B–1. *Saturnian System Statistics (Earth Included for Comparison)*

Object	Diameter or dimensions, km	Density[a]	Gravity	Albedo	Distance from Saturn, km		Orbital period, hr	Closest approach, km		Discoverer	Year of discovery
					Minimum	Maximum		Voyager 1	Voyager 2		
Earth	[b]12 756 [c]12 713	5.5	1.0								
Saturn	[b]120 000 [c]107 000	.7	.9								
Ring D					67 000	73 200	4.91				
Ring C					73 200	92 200	5.61			Galle	1838
Ring B					92 200	117 500	7.93			Huygens	1655
Cassini Division					117 500	121 000				Cassini	1675
Ring A					121 000	136 200	11.93			Huygens	1655
Encke Division					133 500	133 700	13.82			Encke	1837
1980S28	30				137 300		14.45	219 000	287 000		
1980S27	220				139 400		14.71	300 000	247 000		
Ring F					140 600		14.94				
1980S26	200				141 700		15.08	270 000	107 000		
1980S3	90 × 40				151 422		16.66	121 000	147 000		
1980S2	100 × 90				151 472		16.67	297 000	223 000		
Ring G					170 000		19.90				
Mimas	394	1.4	.005	.6	188 224	184 440	23.14	88 400	309 990	W. Herschel	1789
Ring E					210 000	300 000	27.30				
Enceladus	502	1.1		1.0	240 192		33.36	202 040	87 140	W. Herschel	1789
1980S13	34 × 28 × 26				294 700		45.31				
1980S25	34 × 22 × 22				294 700		45.31				
Tethys	1 048	1.0		0.8	296 563		45.76	415 670	93 000	Cassini	1684
1980S6	160				378 600		65.73	230 000	270 000		
Dione	1 118	1.4		0.6	379 074		66.13	161 520	502 250	Cassini	1684
Rhea	1 528	1.2		0.6	527 828		108.66	73 980	645 280	Cassini	1672
Titan	5 150	1.9			1 221 432		382.50	6 490	665 960	Huygens	1655
Hyperion	205 × 130 × 110			0.3	1 502 275		521.74	880 440	470 840	Bond and Lassell	1848
Iapetus	1 448	1.2		0.5	3 559 400		1901.82	2 470 000	909 070	Cassini	1671
Pheobe	220				10 583 200		9755.67	13 537 000	1 473 000	Pickering	1899

[a]Water = 1.
[b]Equatorial.
[c]Polar.
Sources: Stone and Miner, 1981; Smith et al., 1981; Collins et al., 1980; Davies, 1983.

bodies, larger than the outer irregular satellites of Jupiter and five are larger than the largest asteroid, Ceres.

The sizes and other statistics regarding objects in the Saturnian system are listed in table B–1 and further illustrated in figures 3 (Introduction) and B–1. Comparison of the total known solid surface area in the Saturnian system with that in the whole solar system is illustrated in figure 2.

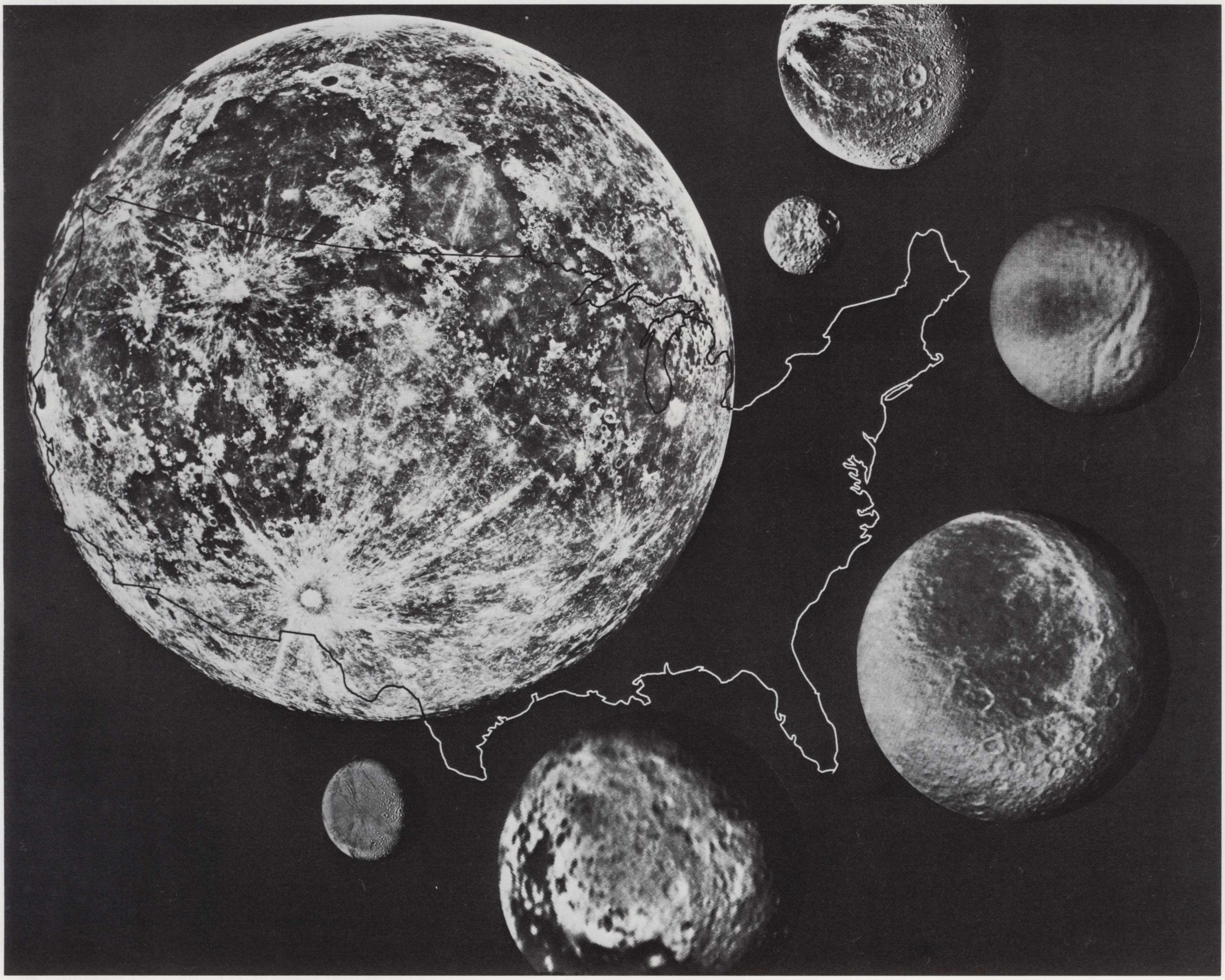

Figure B–1. Relative sizes of six Saturnian satellites. The Moon and an outline of the United States are included for scale.

APPENDIX C

Image Processing

The Voyager pictures of the Saturnian satellites are digital images; they are transmitted from the spacecraft as a stream of numbers, each representing a shade of gray of a specific point in the picture. When these numbers are placed in their proper location in an array of rows and columns, and each number replaced by a tone of the correct gray shade, an image is formed that is visible to the human observer (fig. C–1). These operations are performed with computers and computer-driven film-writing devices. The technique is called "digital image processing." (See Moik, 1980.) Although the methods were originally developed by space scientists for processing lunar pictures, they are now used in medicine for enhancing X-ray photographs, by resource scientists for enhancing aerial and spacecraft views of Earth, and by a variety of other specialists.

Two levels of digital processing are commonly used for planetary mapping. Level 1 is intended to remove all image artifacts, including noise and shading. Level 2 processing changes the geometric shape of the image to match an appropriate map projection. The level 1 and 2 images are carefully preserved on magnetic tape without contrast enhancement. Contrast enhancements are then applied to film images as needed, without modifying the digital tapes. Each Voyager picture contains 800 rows of 800 picture elements, called "pixels" (fig. C–2). Each pixel is assigned a density number (DN) by the spacecraft imaging system, according to the brightness of an image projected on the vidicon tube by the Voyager camera. (See app. A.) The Voyager imaging system is capable of discriminating and transmitting 256 shades of gray. The DNs in a Voyager picture thus range from 0 (black) to 255 (white). The camera can be commanded to take pictures at shutter speeds ranging from 1/10th to 1/200th of a second (table A–1).

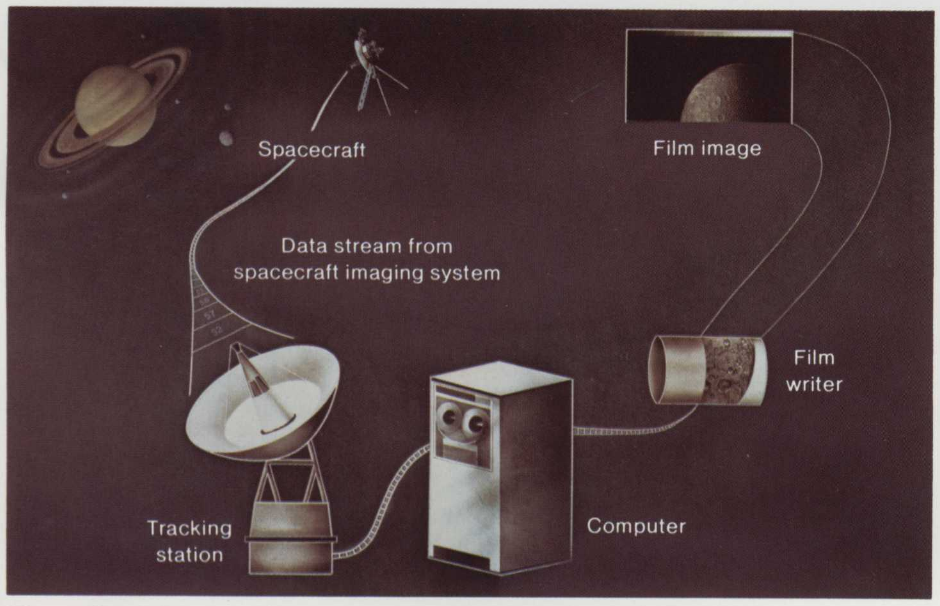

Figure C–1. The principle of the digital image. A stream of numbers (density numbers (DNs)) representing shades of gray is received from the spacecraft by tracking stations on Earth. Each DN is placed in its correct location in an array of rows and columns ("raster") by a computer and is then replaced by its correct gray shade and printed on film by a computer-driven film-writing device. (Illustration by Susan L. Davis)

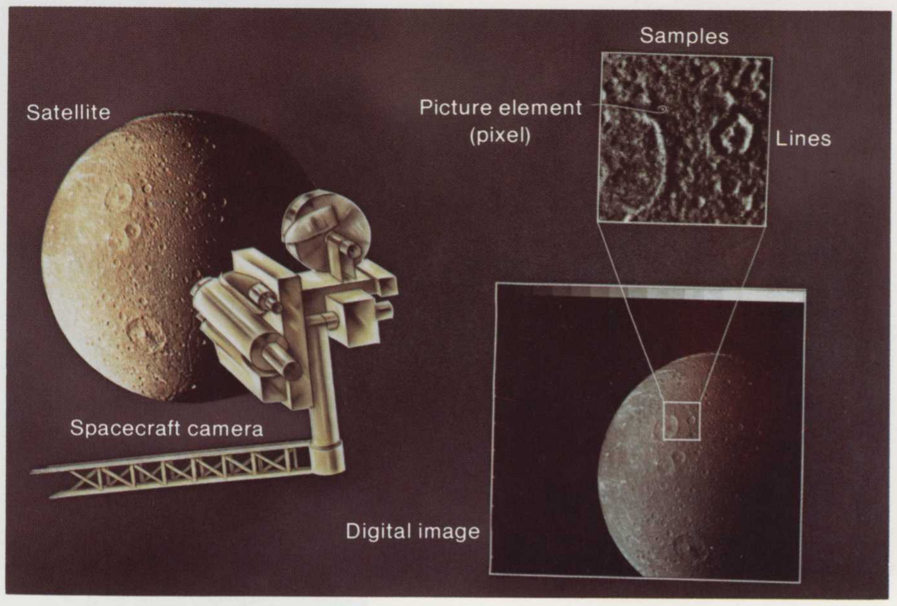

Figure C–2. Spacecraft digital imaging system. Picture elements, or pixels are visible on the enlarged inset. (Illustration by Patricia M. Bridges)

A variety of factors degrade any spacecraft television picture. Electronic fluctuations within the spacecraft cause a variation in the sensitivity of the system and make the DN that should represent black, or no signal, larger than 0. Spurious DN values are often injected into an image by fluctuations in electrical currents or magnetic fields in or near the camera. Segments of a string of DN values may be lost during transmission. Although lens distortions are virtually negligible in the Voyager cameras, geometric distortions are introduced into the pictures by the electronic recording and transmitting systems. All of these problems can be corrected, or significantly reduced, by digital image processing techniques. For example, pictures of empty space will contain many DN values other than 0 in a pattern that does not vary significantly from one frame to the next (fig. C–3). Because it is known that all these values should equal 0, subtracting DN values in black-sky pictures from the DN values in the same rows and columns on pictures of a planet or satellite results in a picture that has been corrected for radiometric distortions.

Spurious DN values (bit errors and dropped lines) can be detected by computer programs that examine the rate of change in DN values that are adjacent to each other in an image. Abrupt changes do not usually occur in images of natural surfaces, but are likely to result from data anomalies. They are corrected by replacing the DN values of pixels that deviate significantly from their surroundings with the average DN value of neighboring pixels.

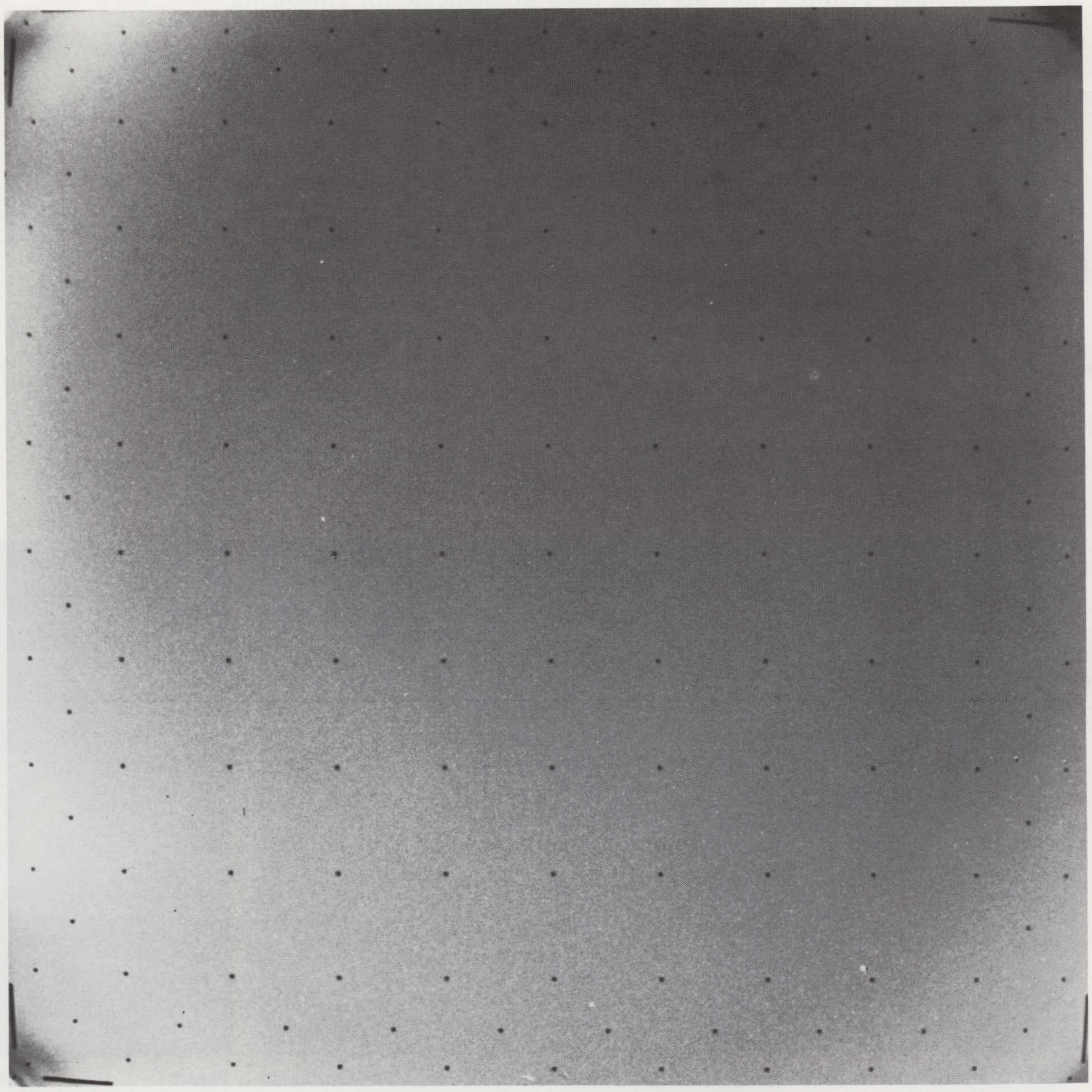

Figure C–3. A picture of black, featureless space used for in-flight calibration of Voyager cameras. Nonblack tones (exaggerated in this illustration for emphasis) are artifacts of the spacecraft imaging system. The DNs in this image are subtracted from raw images to produce radiometrically corrected pictures. Picno 0998J1–025.

The purpose of level 1 processing is to restore the pictures to the quality that would have been produced by a "perfect" camera transmission and film recording system. Figures C–4 and C–5 illustrate this processing phase. Figure C–4 is a "raw," or uncorrected, Voyager picture of Dione. Figure C–5 is the same picture after radiometric correction, reseau and blemish removal, and contrast enhancement. This atlas is intended to show the preliminary cartographic products of the Voyager mission, and hence contains only level 1 images.

Final map compilations are being done with level 2, or geometrically processed, images. This processing entails removal of camera distortions and transformation of the images to appropriate map projections. Distortion corrections are based on the known positions of calibration dots (reseau marks) in an image. These marks are etched in the photosensitive surface on the vidicon faceplate so that they appear in every picture. The positions of the reseau marks were measured on Earth before the spacecraft was launched. When the pictures were received on Earth, the reseau marks were located in each image and their positions recorded. The reseau pattern is clearly visible in figures C–3 and C–4. Although the marks themselves are removed during the level 1 phase by processing similar to that used for bit errors and dropped lines, their recorded positions can still be used to control geometric transformations. In correcting distortions, each image is stretched like a rubber sheet in such a way as to restore the reseau marks to their original, correct locations.

Each geometric correction of a digital image results in a slight loss in resolution. For this reason the correction of camera distortions is done

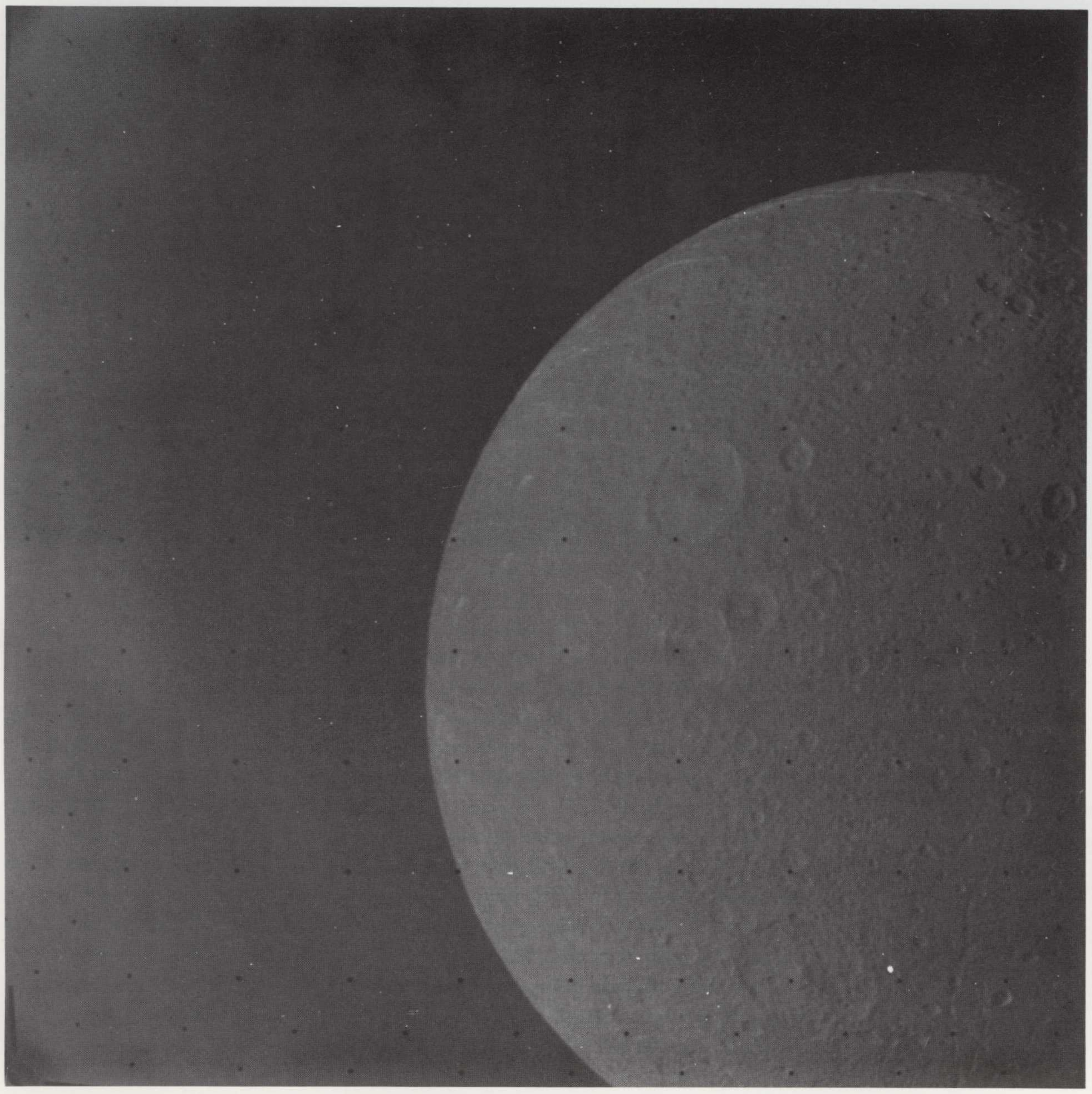

Figure C–4. An untreated Voyager picture of Dione. Picno 0272S1+000.

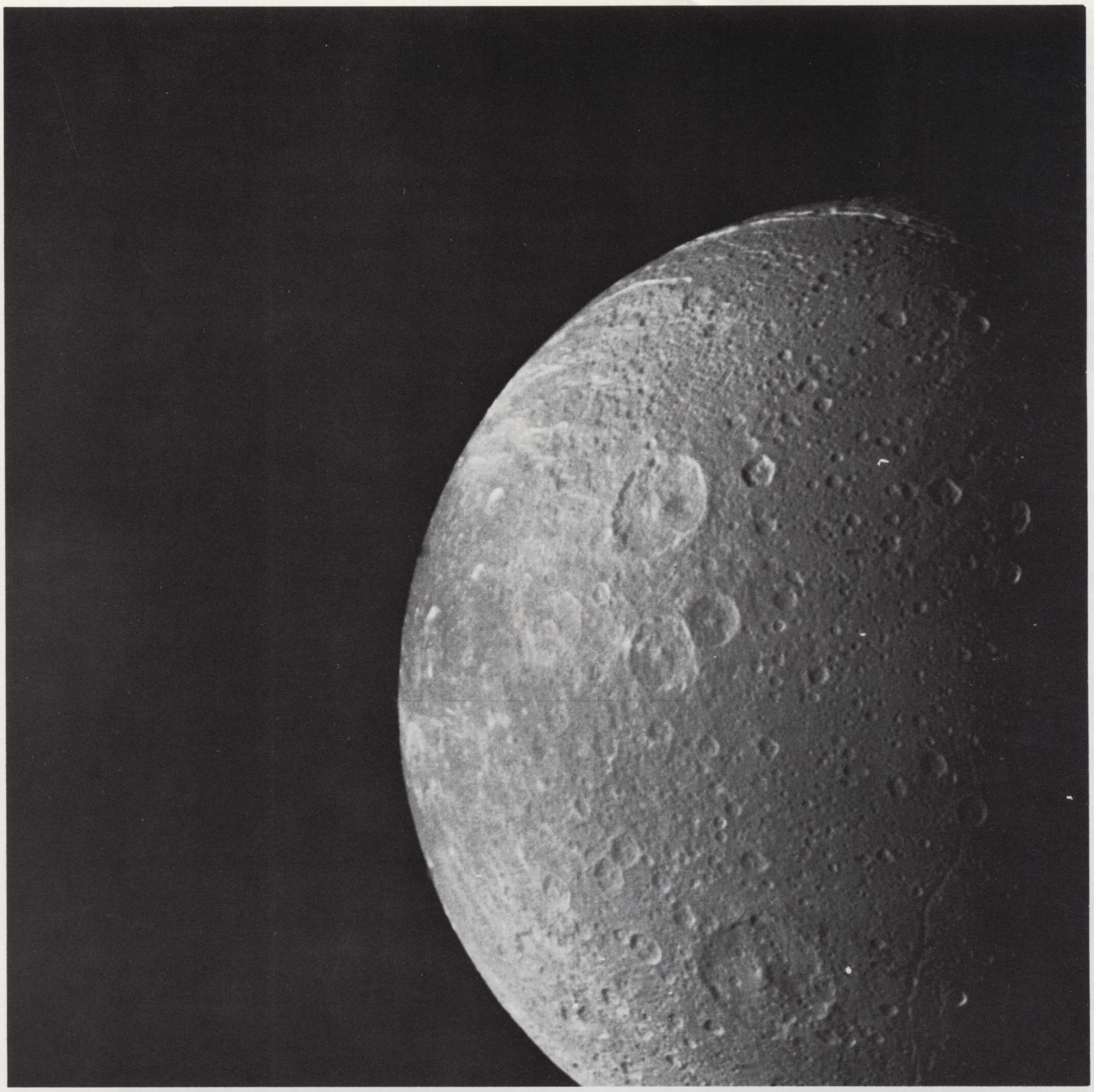

Figure C–5. Voyager picture of Dione with bit errors, dropped lines, and reseau marks removed and contrast enhanced (level 1 processing). Picno 0272S1+000.

simultaneously with transformation to a map projection, so that the actual geometric correction is performed only once.

Transformation to map projections is based on orientation matrices derived by analytical photogrammetry (Davies and Katayama, 1983*a,b*). This process involves a complex mathematical analysis of the positions of images of selected features (control points) on several spacecraft television pictures. The final results of this calculation include precise latitudes and longitudes for the control points and a set of linear equations that precisely define the orientation of the spacecraft camera with respect to the satellite at the time each picture was taken. These equations are used to change the shape of mapping pictures so that they will fit specified projections. Figure C–6 is a geometrically corrected, map-projected version of the Dione image of figure C–5.

A digital television image, with its 256 shades of gray, frequently contains far more information than can be discriminated by the human eye or shown on film. Before it is printed on film (with or without geometric correction), a decision must be made as to which aspect of the image to emphasize. If it contains both very dark and very light areas, its contrast must be modified in the computer so that detail in both light and dark areas will be visible.

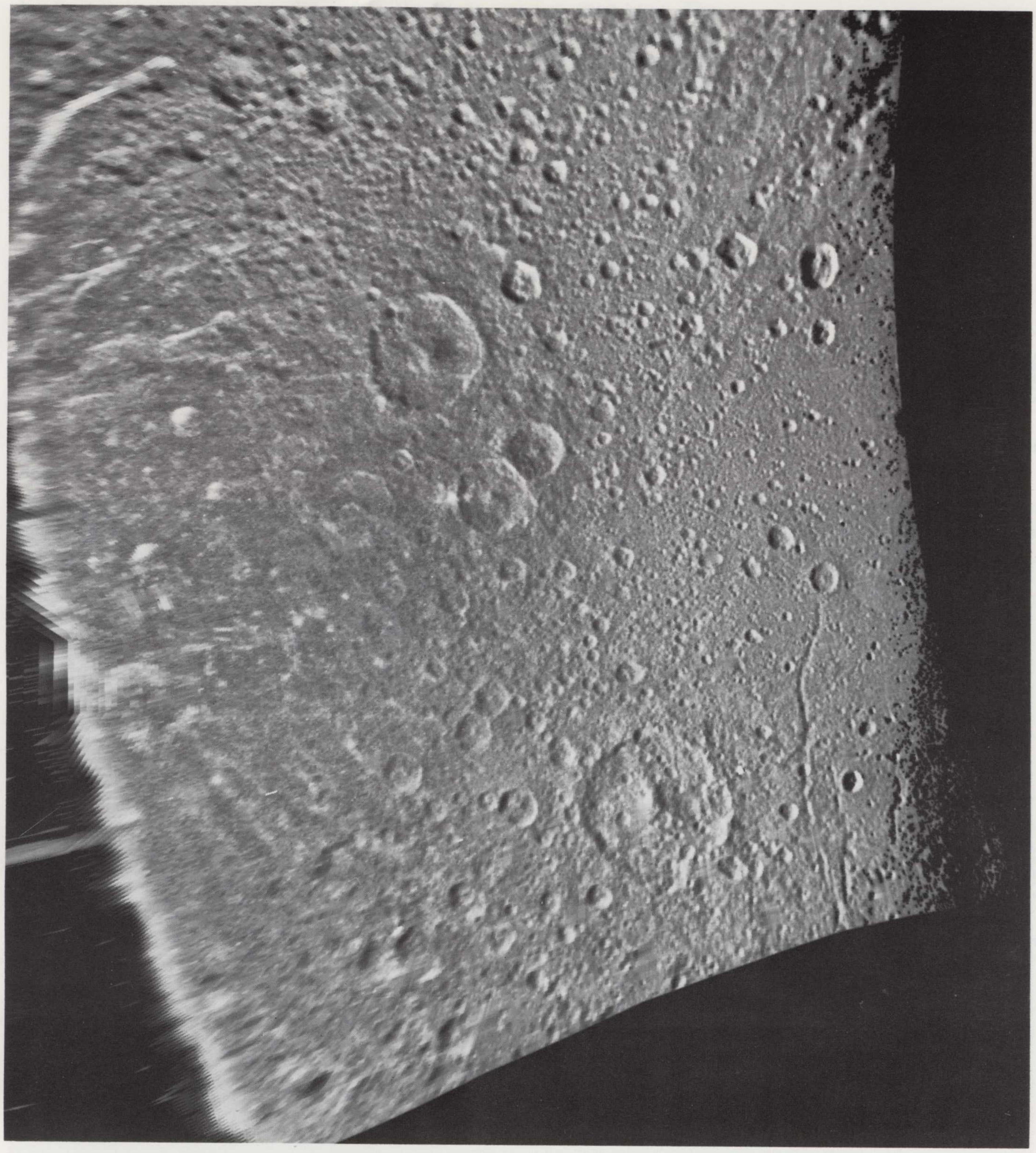

Figure C–6. Geometrically corrected Mercator projection of figure C–5 (level 2 processing). High-pass filter and contrast stretch have been applied to this image.

Two kinds of contrast manipulation are commonly used either singly or in combination on digital images. The first is a simple contrast change ("stretch") applied to the whole image. This is analogous to the photographic processing operation in which photographic papers of different contrast grades are used.

Contrast-stretch parameters for a given image are selected on the basis of a histogram of DN values in the image (figure C–7). This histogram is a graph that shows the number of pixels of each DN value in the image. The lowest DN recorded for a significant number of pixels in the original image is set to 0 for the stretched image. The highest DN recorded for a significant number of pixels in the original image is set to 255 DN for the stretched image. Intermediate DNs are reset in proportion to the newly defined high and low values. Figures C–4 and C–5 illustrate the effect of contrast stretching.

The second method of contrast enhancement, called high-pass filtration, changes local contrasts throughout the picture in such a way that small

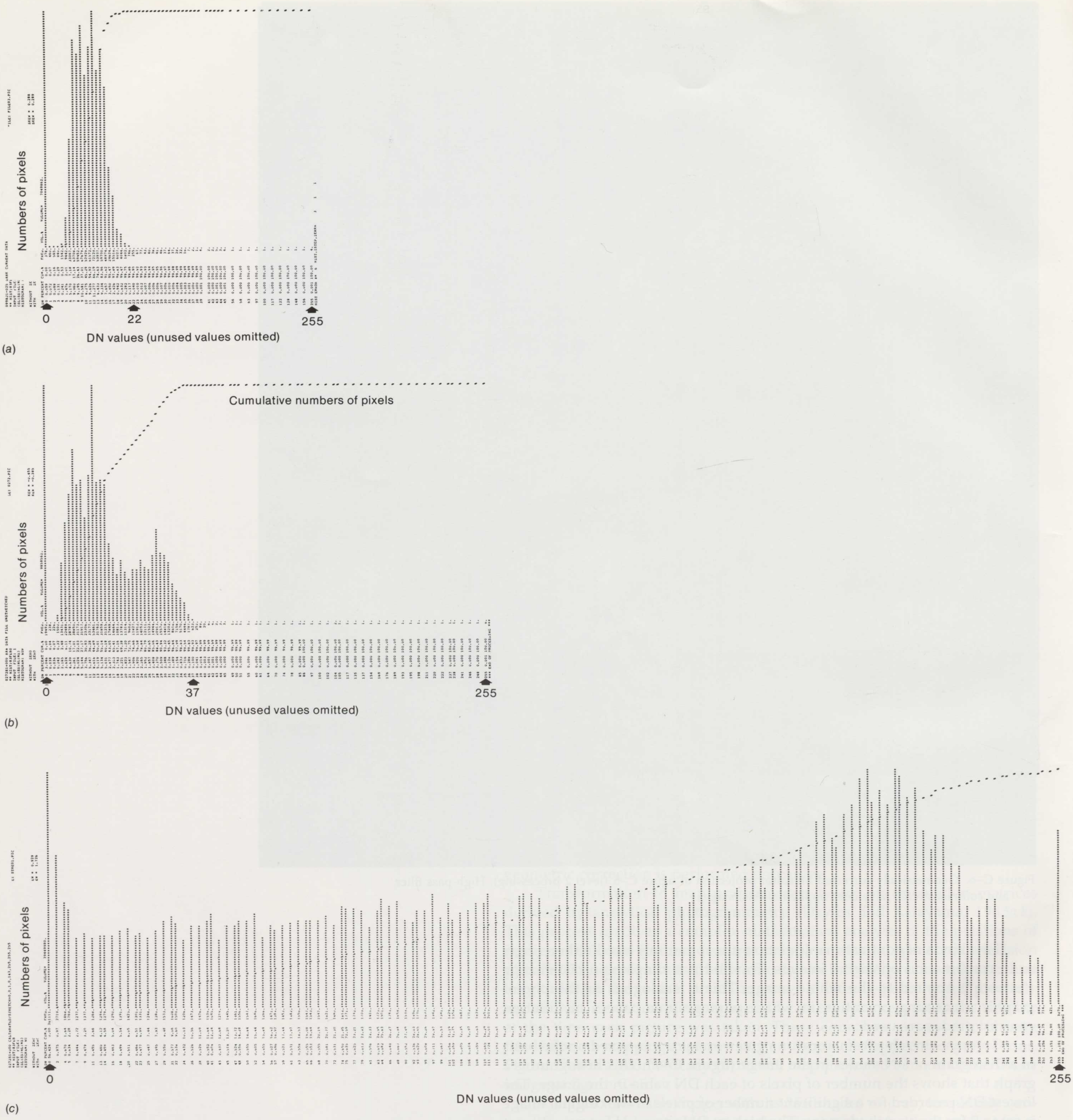

Figure C–7. Histograms of DNs in picno 0272S1+000. (*a*) The calibration image of figure C–3. (*b*) The raw, untreated version. Note that all DN values are clustered in a 37-DN range, including the DNs in (*a*), representing black sky. This is the histogram for the image of figure C–5. (*c*) The contrast-stretched version of the level 1 image in which the black-sky frame has been subtracted and the DNs in the raw image with values of 37 or greater have been converted to 255. Intermediate values are modified in proportion. This is the histogram for the image of figure C–8.

details, bright or dark, are adjusted to the same average gray tone. The effect of this computer technique is analogous to that of electronic dodging devices available on some photographic printers and enlargers, or the photographic technique of "unsharp masking." The small details are emphasized while broader tonal variations in the picture are subdued (figs. C–5 and C–8).

The following procedure is used to make high-pass filtered versions of digital images (fig. C–9):

(1) The filter size is selected on the basis of the size of image features to be emphasized. This size is defined in terms of pixel dimensions: a filter might be 3 pixels wide by 3 pixels high, or 51 pixels wide by 31 pixels high. The filter dimensions are odd numbers so that there is always a pixel in the exact center of the filter.

(2) The high-pass filter computer program computes the average DN value of all pixels within the filter beginning at its first location in the upper left corner of the picture. The actual DN of the pixel in the upper left corner of the filter box is subtracted from this average. Because there is about a 50-percent chance that this new filtered

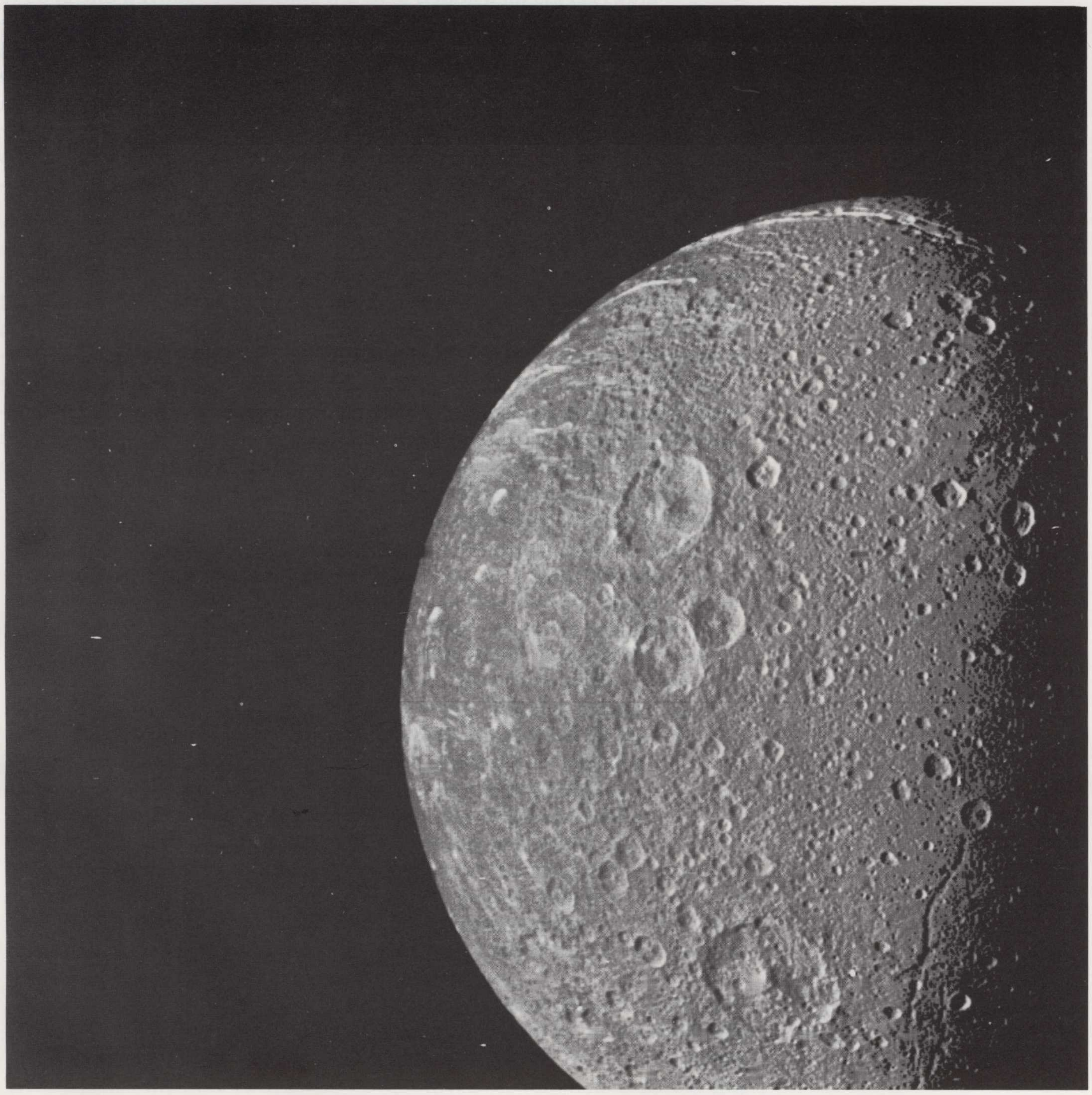

Figure C–8. High-pass-filtered version of figure C–5. A 101-×-101-pixel filter (300 × 300 km on Dione) was used on this image, thus subduing tonal variations covering areas larger than 300 × 300 km. Note also that midtones are uniform across the image, even near the terminator where the unfiltered image of figure C–5 is much darker.

56	37	32	10	15	48	40
53	44	38	18	15	30	32
60	55	43	10	14	28	32
75	62	60	61	8	16	25
70	65	54	48	32	18	20
65	71	66	55	46	38	41
51	58	59	54	49	41	43

(*a*)

x	x	x	x	x	x	x
x	46	32	22	21	29	x
x	54	43	30	22	23	x
x	60	38	23	26	21	x
x	65	60	48	36	27	x
x	62	59	51	42	36	x
x	x	x	x	x	x	x

(*b*)

x	x	x	x	x	x	x
x	125	133	123	121	128	x
x	128	127	107	119	132	x
x	129	149	165	109	122	x
x	127	121	127	123	118	x
x	136	134	131	131	129	x
x	x	x	x	x	x	x

(*c*)

Figure C–9. The arithmetic of high-pass filtration. (*a*) An array of DNs representing a hypothetical raw image seven pixels wide and seven pixels long. (*b*) 3-X-3-pixel low-pass filter image of (*a*). The average DN value of the 3 X 3 block of pixels in the upper left corner of the image was calculated, and the result used as the value of the pixel in the center of the 3-X-3-pixel block. The block was then moved one pixel to the right and the process repeated until the entire image, except for the edges, was filled with the average values for the blocks. Because the pixels at the edge of the image are not at the center of any blocks, they are either lost or treated separately. The effect of this process is to defocus the image. The larger the filter, the greater the defocussing effect. (*c*) The 3-X-3-pixel high-pass filter image of (*a*). Each value in the low-pass filter picture (*b*) is subtracted from its corresponding value in the raw image (*a*) and added to 127, so that negative values do not result and so that the midtone of the image will have an average value of 127, in the middle of the 255-DN tonal range.

DN will have a negative value, a DN value representing a medium gray (127) is then added to it. The filter is then moved one pixel to the right, and the process is repeated. When filtered values for the first row of pixels have been computed, the filter box is moved down one row and filtered values for the second row of pixels are computed. The process is repeated until new values have been computed for each pixel in the image. The filters are affected by frame edges and abrupt contrast changes, so special program steps must be devised to deal with these problems. Several high-pass filter algorithms are in use, but they produce similar results.

Filter sizes for the Voyager pictures in this atlas were selected primarily on the basis of image scale so that landforms would be uniformly emphasized. In other words, rather than using, for example, a 101-X-101-pixel filter for all pictures, a 21-X-21-km filter might be used. To achieve this uniformity, a picture taken when Voyager was 22 500 km from a satellite might be treated with a 101-X-101-pixel high-pass filter, whereas a picture taken from five times farther away (112 000 km) might have a 21-X-21-pixel filter applied. No single filter size, however, was appropriate for the full range of available image resolutions. It was, therefore, necessary to group the images as to high, medium, and low resolution and select the best filter for each group by trial and error.

A Catalog of Selected king Orbiter Images

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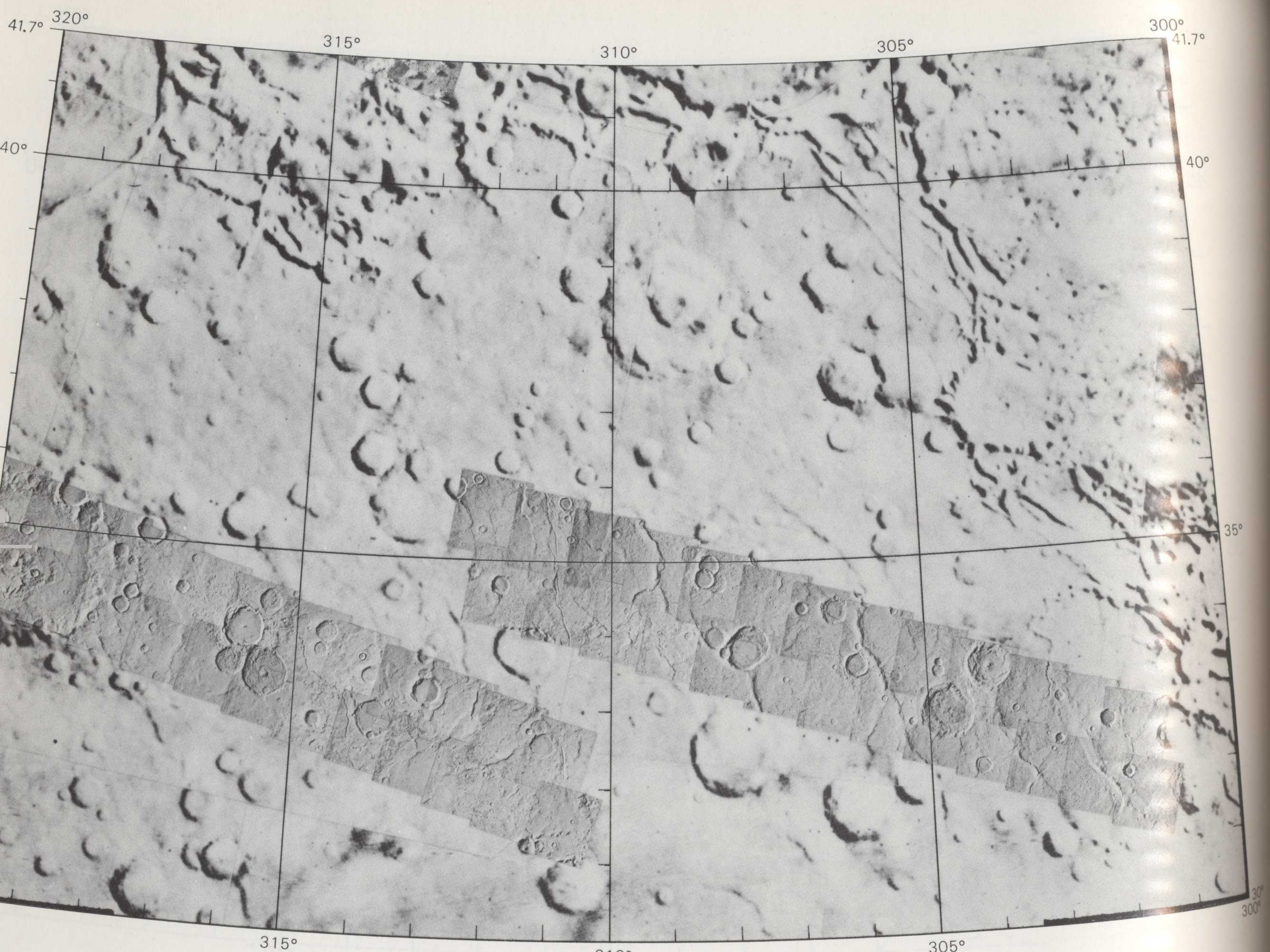
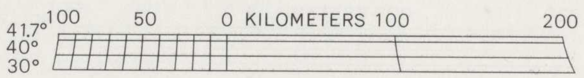
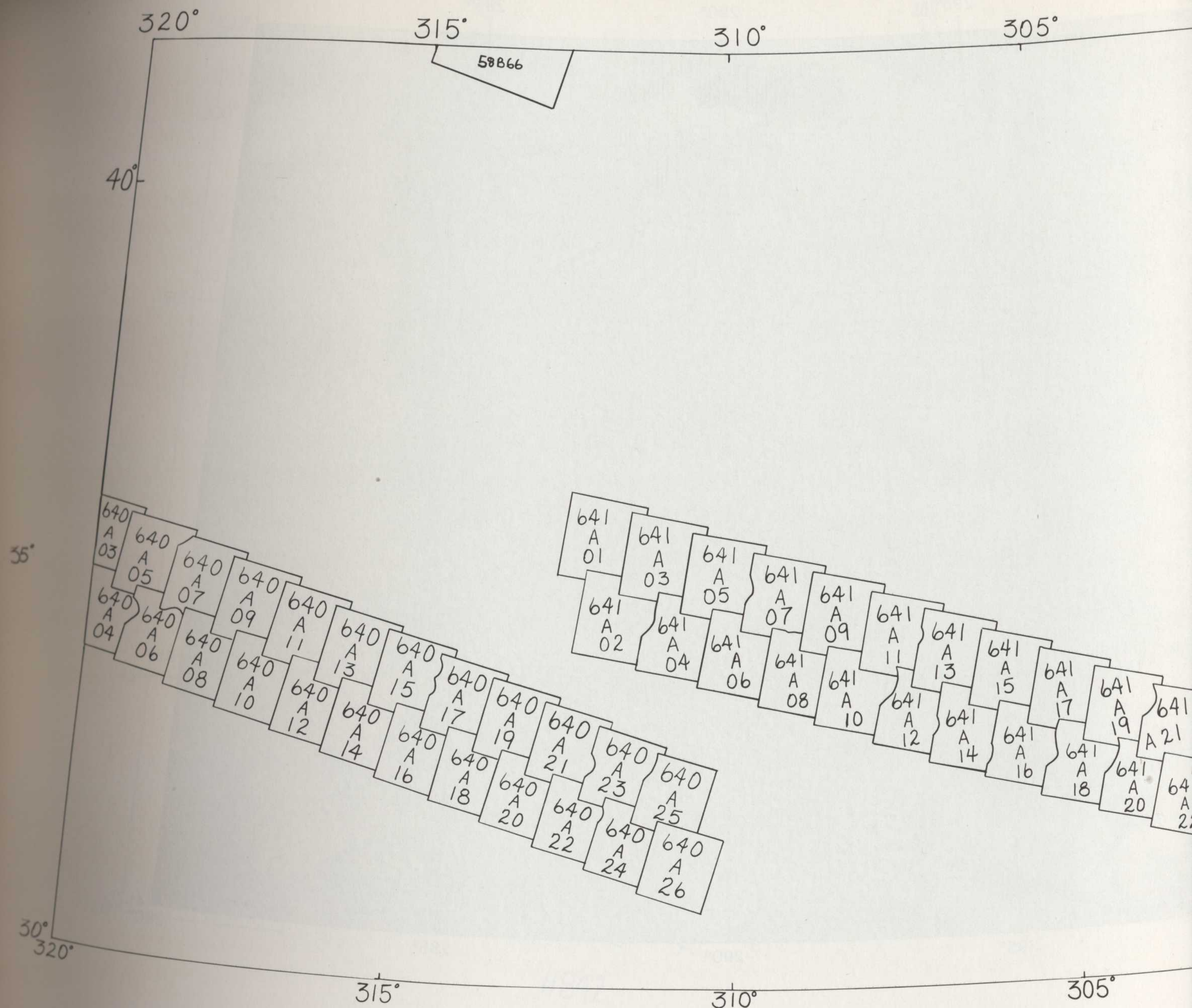


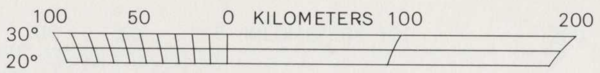


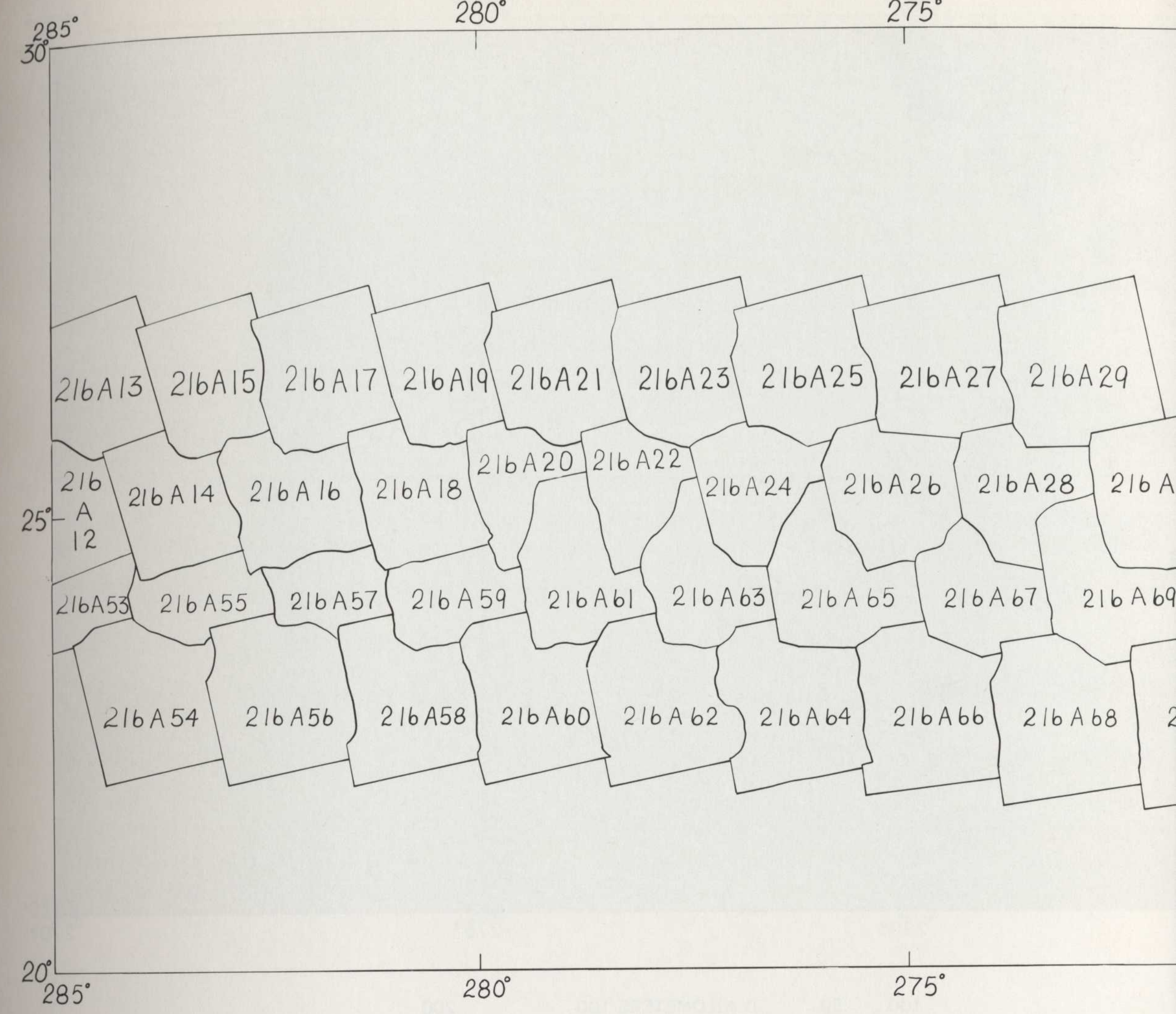
5-SE—Southeast subquad of
e Ismenius Lacus quadrangle.
le 1:4,336,000 at 30° N. lat.
talog photomosaic, above; index
Viking pictures, facing page,
; listing of Viking pictures
ilable as of November 1978,
ing page, bottom.



MC 5SE VIKING PHOTO COVERAGE AS OF 30-NOV-78
HIGH RESOLUTION >35<180M/PXL

PICNO	FSC	CLAT	CLON	EMA	RMAG	M/PIX
569A 7	36828000	38.24	319.63	7.	9682.	161.
569A 8	36828001	35.52	317.60	3.	9673.	160.
569A 9	36828002	38.29	316.87	7.	9664.	161.
569A10	36828003	35.60	315.06	5.	9656.	159.
569A11	36828004	38.35	314.15	8.	9647.	160.
569A12	36828005	35.53	312.39	8.	9639.	159.
569A13	36828006	38.25	311.40	11.	9630.	160.
569A14	36828007	35.52	309.73	11.	9621.	159.
569A15	36828008	38.22	308.65	13.	9613.	160.
569A16	36828009	35.35	307.13	14.	9604.	159.
569A17	36828010	38.05	305.92	16.	9596.	160.
569A18	36828011	35.24	304.50	17.	9587.	160.
569A19	36828012	37.91	303.18	19.	9578.	160.
569A20	36828013	34.99	301.71	20.	9570.	161.
569A29	36828048	32.19	318.75	5.	9267.	149.
569A31	36828050	32.31	316.36	4.	9249.	149.
569A33	36828052	32.40	313.97	5.	9232.	149.
569A35	36828054	32.39	311.57	7.	9214.	149.
569A37	36828056	32.38	309.20	10.	9197.	149.
569A39	36828058	32.38	306.78	13.	9179.	149.
640A 5	38196298	35.22	319.94	5.	5482.	53.
640A 6	38196299	34.26	319.71	4.	5473.	53.
640A 7	38196300	34.99	319.06	4.	5464.	53.
640A 8	38196301	34.04	318.83	3.	5455.	52.
640A 9	38196302	34.75	318.20	3.	5446.	52.
640A10	38196303	33.81	318.01	1.	5437.	52.
640A11	38196304	34.50	317.33	2.	5428.	52.
640A12	38196305	33.57	317.12	0.	5419.	52.
640A13	38196306	34.29	316.63	2.	5411.	51.
640A14	38196307	33.32	316.35	1.	5402.	51.
640A15	38196308	34.01	315.70	3.	5393.	51.
640A16	38196309	33.09	315.51	2.	5384.	51.
640A17	38196310	33.76	314.86	4.	5375.	51.
640A18	38196311	32.83	314.70	4.	5366.	50.
640A19	38196312	33.49	314.05	5.	5358.	50.
640A20	38196313	32.57	313.89	6.	5349.	50.
641A 9	38215570	34.79	309.14	3.	5462.	53.
641A10	38215571	33.84	308.93	2.	5453.	52.
641A11	38215572	34.53	308.23	2.	5444.	52.
641A12	38215573	33.60	308.07	0.	5436.	52.
641A13	38215574	34.29	307.41	2.	5427.	52.
641A14	38215575	33.35	307.24	1.	5418.	51.
641A15	38215576	34.03	306.56	3.	5409.	51.
641A16	38215577	33.10	306.37	2.	5400.	51.
641A17	38215578	33.76	305.69	3.	5391.	51.
641A18	38215579	32.84	305.55	3.	5382.	51.
641A19	38215580	33.51	304.91	4.	5374.	51.
641A20	38215581	32.59	304.74	5.	5365.	50.
641A21	38215582	33.23	304.12	6.	5356.	50.
641A22	38215583	32.32	303.96	6.	5347.	50.
641A23	38215584	32.96	303.31	7.	5338.	50.
641A24	38215585	32.06	303.17	7.	5329.	50.
641A25	38215586	32.69	302.53	8.	5321.	50.
641A26	38215587	31.79	302.41	8.	5312.	49.



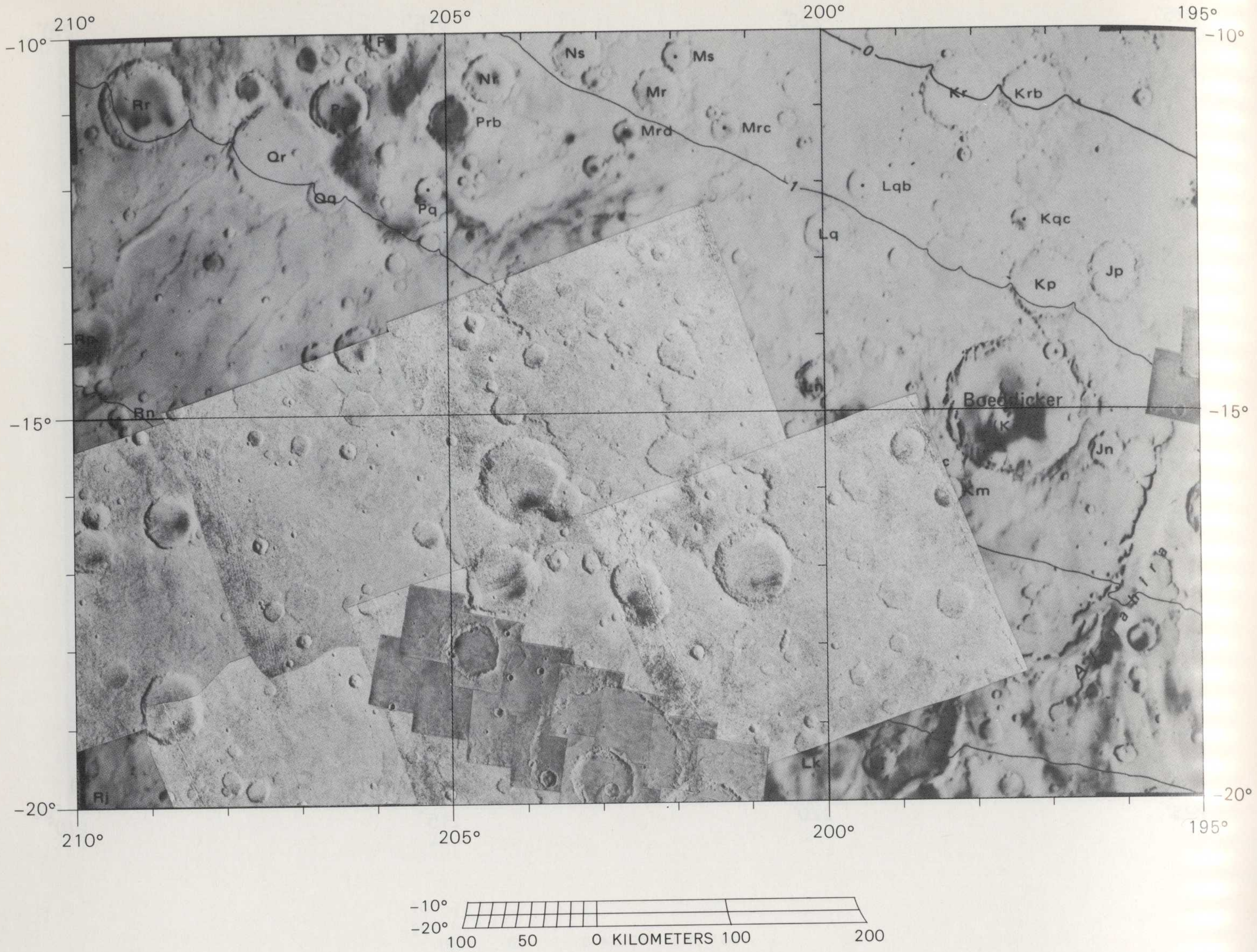


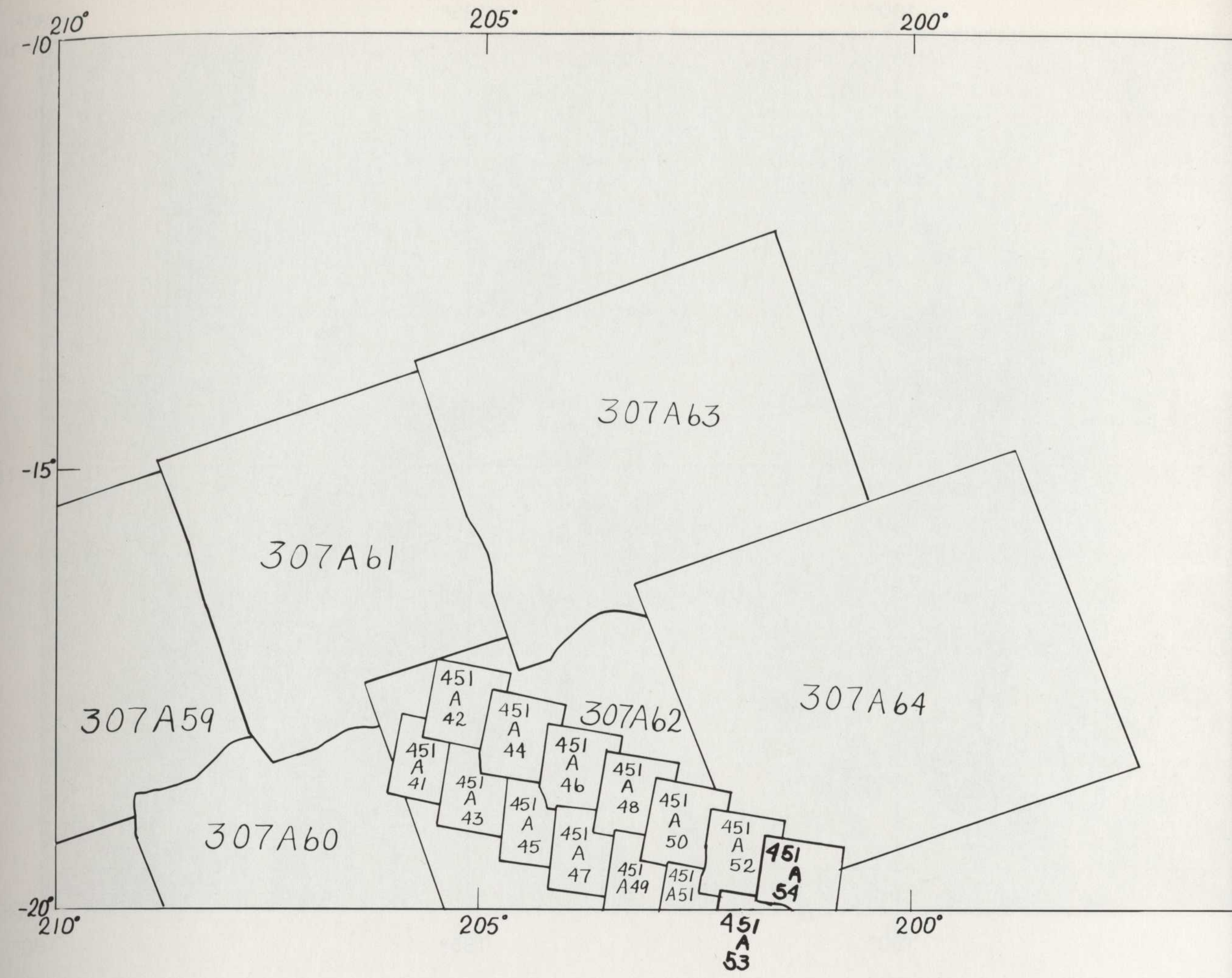
MC-13-NE—Northeast subquad of the Syrtis Major quadrangle. Scale 1:5,000,000 at 0° lat. Catalog photomosaic, above; index of Viking pictures, facing page, top; listing of Viking pictures available as of November 1978, facing page, bottom.

MC13NE VIKING PHOTO COVERAGE AS OF 30-NOV-78
HIGH RESOLUTION >35<180M/PXL

PICNO	FSC	CLAT	CLON	EMA	RMAG	M/PIX
216A12	30128033	24.95	284.79	6.	6420.	77.
216A13	30128034	26.21	284.46	9.	6427.	78.
216A14	30128035	24.98	283.46	8.	6434.	78.
216A15	30128036	26.21	283.18	10.	6441.	79.
216A16	30128037	25.01	282.13	10.	6447.	79.
216A17	30128038	26.24	281.83	12.	6454.	80.
216A18	30128039	25.01	280.78	12.	6461.	80.
216A19	30128040	26.28	280.38	14.	6468.	81.
216A20	30128041	25.03	279.42	14.	6475.	81.
216A21	30128042	26.29	279.03	16.	6482.	82.
216A22	30128043	25.00	277.99	16.	6488.	82.
216A23	30128044	26.30	277.53	18.	6495.	84.
216A24	30128045	24.99	276.59	18.	6502.	84.
216A25	30128046	26.26	276.15	20.	6509.	85.
216A26	30128047	24.96	275.13	20.	6516.	85.
216A27	30128048	26.28	274.62	22.	6523.	87.
216A28	30128049	24.92	273.61	23.	6530.	86.
216A29	30128050	26.27	273.09	24.	6537.	88.
216A30	30128051	24.91	272.04	25.	6543.	88.
216A53	30128094	24.35	284.85	13.	6845.	90.
216A54	30128095	22.93	283.81	9.	6852.	89.
216A55	30128096	24.35	283.38	12.	6860.	90.
216A56	30128097	22.89	282.32	9.	6867.	90.
216A57	30128098	24.35	281.90	12.	6874.	91.
216A58	30128099	22.90	280.85	9.	6881.	90.
216A59	30128100	24.33	280.33	12.	6888.	92.
216A60	30128101	22.86	279.31	10.	6895.	91.
216A61	30128102	24.32	278.85	13.	6903.	92.
216A62	30128103	22.84	277.80	11.	6910.	92.
216A63	30128104	24.30	277.27	14.	6917.	93.
216A64	30128105	22.79	276.31	12.	6924.	93.
216A65	30128106	24.26	275.72	15.	6931.	94.
216A66	30128107	22.74	274.72	14.	6938.	94.
216A67	30128108	24.22	274.17	17.	6946.	96.
216A68	30128109	22.67	273.14	16.	6953.	95.

534A55	36153772	28.01	275.58	20.	8955.	145
534A56	36153773	25.67	274.86	23.	8946.	14
572A22	36885863	27.97	275.49	18.	8841.	147
572A24	36885865	27.91	273.24	21.	8823.	14
572A26	36885867	27.79	270.96	23.	8805.	14

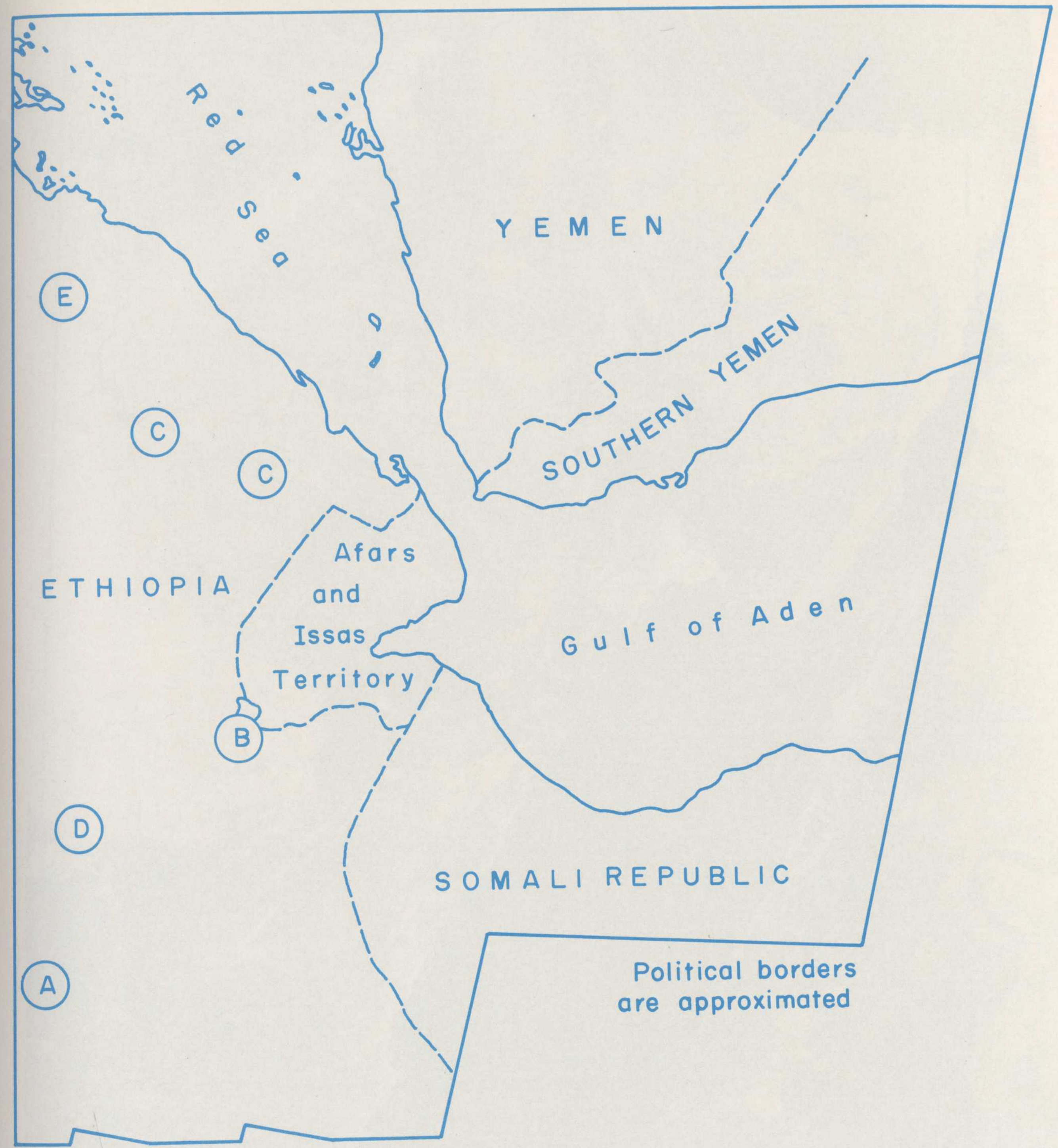




MC23 C VIKING PHOTO COVERAGE AS OF 30-NOV-78
HIGH RESOLUTION >35<180M/PXL

PICNO	FSC	CLAT	CLON	EMA	RMAG	M/PIX
451A41	34555696	-18.01	205.86	16.	5210.	48.
451A42	34555697	-17.41	205.44	16.	5218.	48.
451A43	34555698	-18.35	205.21	15.	5227.	48.
451A44	34555699	-17.75	204.80	15.	5236.	48.
451A45	34555700	-18.68	204.60	14.	5245.	48.
451A46	34555701	-18.08	204.15	14.	5253.	48.
451A47	34555702	-19.01	203.97	13.	5262.	48.
451A48	34555703	-18.40	203.51	12.	5271.	49.
451A49	34555704	-19.33	203.34	12.	5279.	49.
451A50	34555705	-18.72	202.88	11.	5288.	49.
451A51	34555706	-19.67	202.69	10.	5297.	49.
451A52	34555707	-19.04	202.24	10.	5306.	49.
451A53	34555708	-20.00	202.04	9.	5314.	49.
451A54	34555709	-19.37	201.60	9.	5323.	49.
451A56	34555711	-19.68	200.93	7.	5341.	50.
451A58	34555713	-20.00	200.28	6.	5358.	50.
452A 1	34574946	-14.56	195.62	15.	5104.	45.
452A 2	34574947	-13.99	195.22	15.	5112.	46.
748A61	40277980	-19.14	207.27	1.	8644.	133.
748A62	40277981	-19.04	207.25	1.	8654.	133.
748A63	40277982	-17.93	207.94	3.	8663.	134.
748A64	40277983	-17.86	207.89	3.	8672.	134.
748A65	40277984	-16.76	208.61	5.	8681.	134.
748A66	40277985	-16.67	208.55	5.	8690.	135.

C-23-C—Central subquad of the Aeolis quadrangle. Scale :5,000,000 at 0° lat. Catalog photomosaic, above; index of Viking pictures, facing page, top; listing of Viking pictures available as of November 1978, facing page, bottom.



SOUTHERN RED SEA: The Red Sea, Gulf of Aden, and East African rifts (*A*) (see plate 272 for letter identifications), an important part of the world rift system, meet at the triple junction of the Afar triangle. (See plate 366.) The junction, shown here at a smaller scale, is centered at *B* in the large dark triangular-shaped Afar depression. To the geologist, this scene depicts an area of incipient crustal fragmentation and probable movement of continental plates. That is, the Arabian plate, occupied by Yemen and Southern Yemen on this mosaic, was once a part of Africa. About 35 million years ago, the continental plate ruptured along what is now the approximate axes of the Red Sea and Gulf of Aden, and the Arabian plate moved in a northwesterly direction away from Africa. The Afar thus represents an area in which the process of "sea-floor" spreading is taking place on land; that is, much of the area is exposed "sea floor" and many of the dark volcanic rocks (*C*) are of compositions typical of oceanic basalts. Much of the interior is, in fact, topographically below sea level, with Lake Assale (*D*) in the Danakil depression (*E*) at 116 meters (361 feet) below sea level being the second lowest terrestrial area in the world.

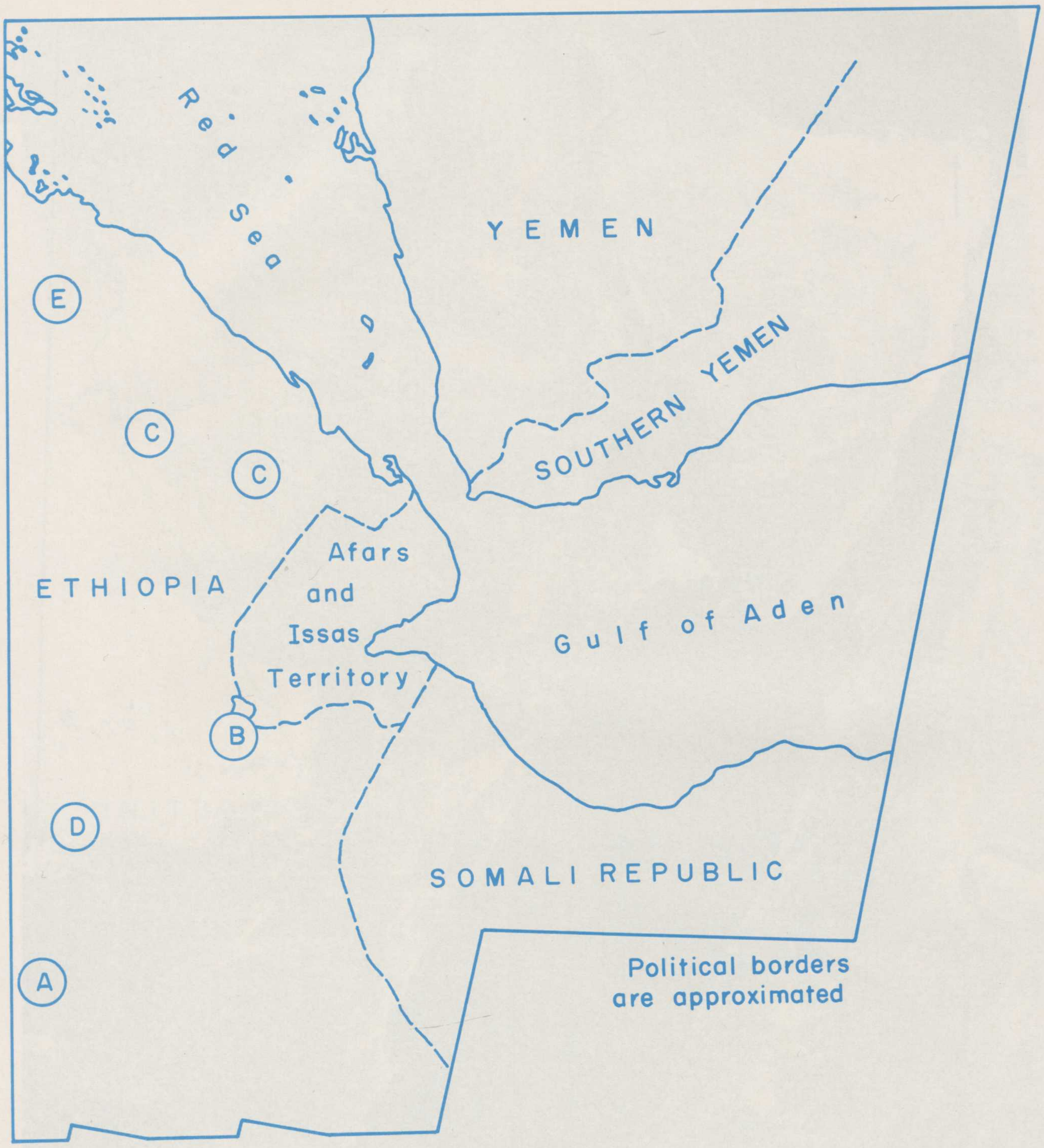
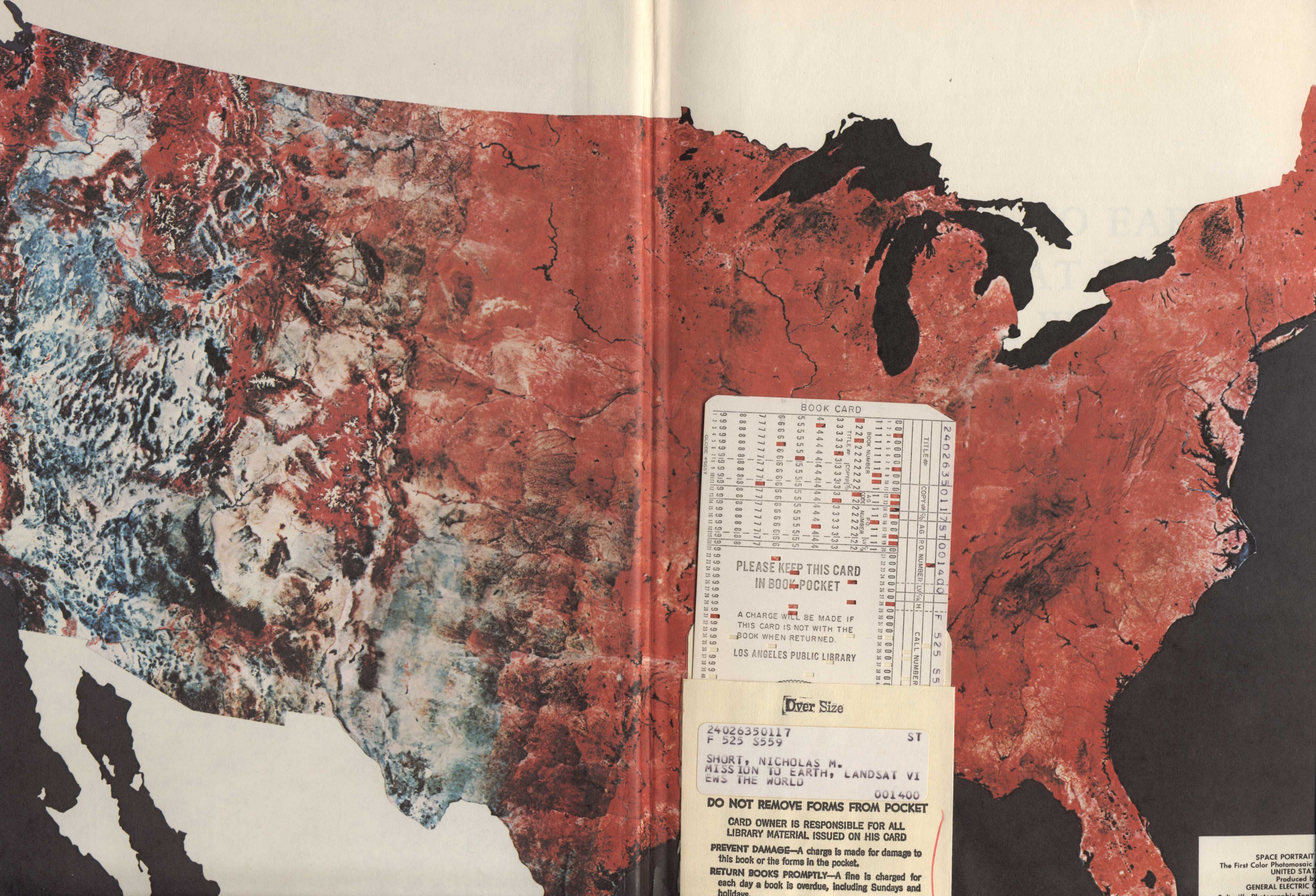

Red Sea
YEMEN
SOUTHERN YEMEN
E
C
C
ETHIOPIA
Afars
and
Issas
Territory
Gulf of Aden
B
D
SOMALI REPUBLIC
A
Political borders
are approximated

PLATE 272

PLATE 271

SOUTHERN RED SEA: The Red Sea, Gulf of Aden, and East African rifts (*A*) (see plate 272 for letter identifications), an important part of the world rift system, meet at the triple junction of the Afar triangle. (See plate 366.) The junction, shown here at a smaller scale, is centered at *B* in the large dark triangular-shaped Afar depression. To the geologist, this scene depicts an area of incipient crustal fragmentation and probable movement of continental plates. That is, the Arabian plate, occupied by Yemen and Southern Yemen on this mosaic, was once a part of Africa. About 35 million years ago, the continental plate ruptured along what is now the approximate axes of the Red Sea and Gulf of Aden, and the Arabian plate moved in a northwesterly direction away from Africa. The Afar thus represents an area in which the process of "sea-floor" spreading is taking place on land; that is, much of the area is exposed "sea floor" and many of the dark volcanic rocks (*C*) are of compositions typical of oceanic basalts. Much of the interior is, in fact, topographically below sea level, with Lake Assale (*D*) in the Danakil depression (*E*) at 116 meters (361 feet) below sea level being the second lowest terrestrial area in the world.



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